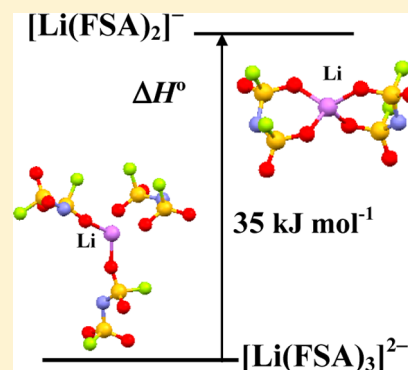


Unusual Li^+ Ion Solvation Structure in Bis(fluorosulfonyl)amide Based Ionic LiquidKenta Fujii,[†] Hiroshi Hamano,[‡] Hiroyuki Doi,[○] Xuedan Song,[§] Seiji Tsuzuki,^{*,||} Kikuko Hayamizu,^{||} Shiro Seki,[⊥] Yasuo Kameda,[#] Kaoru Dokko,[▽] Masayoshi Watanabe,[▽] and Yasuhiro Umebayashi^{*,○}[†]Neutron Science Laboratory, Institute for Solid State Physics, The University of Tokyo, Kashiwa, Chiba 277-8581, Japan[‡]Department of Chemistry, Faculty of Science, Kyushu University, Hakozaki, Higashi-ku, Fukuoka 812-8581, Japan[§]State Key Laboratory of Fine Chemicals, Dalian University of Technology, Linggong Road 2, Dalian 116024, People's Republic of China^{||}National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba Center, Tsukuba, Ibaraki 305-8565, Japan[⊥]Materials Science Research Laboratory, Central Research Institute of Electric Power Industry, Komae, Tokyo 201-8511, Japan[#]Department of Material and Biological Chemistry, Faculty of Science, Yamagata University, Kojirakawa-machi 1-4-12, Yamagata 990-8560, Japan[▽]Department of Chemistry and Biotechnology, Yokohama National University, 79-5 Tokiwadai Hodogaya-ku, Yokohama 240-8501, Japan[○]Graduate School of Science and Technology, Niigata University, 8050, Ikarashi, 2-no-cho, Nishi-ku, Niigata City, 950-2181, Japan

S Supporting Information

ABSTRACT: Raman spectra of 1-ethyl-3-methylimidazolium bis(fluorosulfonyl)amide $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ ionic liquid solutions dissolving LiFSA salt of various concentrations were measured at 298 K. FSA^- ($(\text{FSO}_2)_2\text{N}^-$) is an analogue anion of bis(trifluoromethanesulfonyl)amide ($(\text{CF}_3\text{SO}_2)_2\text{N}^-$; TFSA^-). We found that a solvation number of the Li^+ ion in $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ is 3, though it has been well established that Li^+ ion is solvated by two TFSA^- anions in the corresponding ionic liquids below the Li^+ ion mole fraction of $x_{\text{Li}^+} < 0.2$. To yield further insight into larger solvation numbers, Raman spectra were measured at higher temperatures up to 364 K. The Li^+ ion solvation number in $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ evidently decreased when the temperature was elevated. Temperature dependence of the Li^+ ion solvation number was analyzed assuming an equilibrium between $[\text{Li}(\text{FSA})_2]^-$ and $[\text{Li}(\text{FSA})_3]^{2-}$, and the enthalpy ΔH° and the temperature multiplied entropy $T\Delta S^\circ$ for one FSA^- liberation toward a bulk ionic liquid were successfully evaluated to be $35(2) \text{ kJ mol}^{-1}$ and $29(2) \text{ kJ mol}^{-1}$, respectively. The ΔH° and ΔS° suggest that the Li^+ ion is coordinated by one of bidentate and two of monodentate FSA^- at 298 K, and that the more weakly solvated monodentate FSA^- is liberated at higher temperatures. The high-energy X-ray diffraction (HEXRD) experiments of these systems were carried out and were analyzed with the aid of molecular dynamics (MD) simulations. In radial distribution functions evaluated with HEXRD, a peak at about $1.94 \text{ \AA}$ appeared and was attributable to the $\text{Li}^+-\text{O}(\text{FSA}^-)$ correlations. The longer $\text{Li}^+-\text{O}(\text{FSA}^-)$ distance than that for the $\text{Li}^+-\text{O}(\text{TFSA}^-)$ of $1.86 \text{ \AA}$ strongly supports the larger solvation number of the Li^+ ions in the FSA^- based ionic liquids. MD simulations at least qualitatively reproduced the Raman and HEXRD experiments.



■ INTRODUCTION

Room-temperature ionic liquids have been widely applied to synthetic and separation media as a green chemistry, in particular, to new electrolytes for electrochemical devices such as Li secondary batteries with high safety.^{1–3} For this purpose, it is important to reduce the viscosity of ionic liquids to achieve higher ionic conductivity, so various ionic liquids have been so far proposed as low viscous ones.^{4–10} Bis-(trifluoromethanesulfonyl)amide (TFSA^-) yields a lot of low viscous ionic liquids being combined with various organic cations. TFSA^- based ionic liquids are utilized as electrolytes for Li secondary batteries using ionic liquids, and thus

electrochemical properties for the TFSA^- based ionic liquids containing LiTFSA salt have been well explored.^{11–22} However, there are some technical problems in terms of formation of solid electrolyte interface and/or intercalation to the electrodes in applying to the Li batteries. Bis(fluorosulfonyl)amide (FSA^-) is an analogue anion of TFSA^- , and it has the molecular structure that the CF_3 groups of TFSA^- are substituted by F atoms. FSA^- based ionic liquids may overcome such technical

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problems, and show better performance as an electrolyte for Li batteries.^{23–33} As the FSA[−] based ionic liquids have a lower viscosity relative to the corresponding TFSA[−] based ones, they have also advanced in terms of current density. Other useful applications to electrochemical devices have been also reported such as a field effect transistor^{34,35} and an electric double layer capacitor³⁶ using FSA[−] based ionic liquids.

From a basic physicochemical viewpoint, thermal behaviors of the FSA[−] based ionic liquids with and without a lithium salt have been investigated.^{37–41} For further understanding of macroscopic thermodynamic and transport properties of the FSA[−] ionic liquids, it is indispensable to clarify their macroscopic ones at a molecular level. Diffusion coefficients by the pulse gradient spin echo (PGSE) NMR technique^{42–46} and molecular dynamics simulations^{47,48} have been reported for the FSA[−] based ionic liquids. We have so far studied conformational isomerism and liquid structure in the TFSA[−]^{49–53} and FSA[−]^{54,56} based ionic liquids. Moreover, Li⁺ ion solvation plays an essential role for the performance as batteries, because solvation/desolvation processes are closely related to the whole redox reaction of the Li⁺ ion at the solution/solid interface of the battery electrode. The solvation structure of Li⁺ ion in ionic liquids, particularly in the TFSA[−] based ones, has been well studied by experimental^{57–63} and theoretical^{64–69} techniques. In the TFSA[−] based ionic liquids, it has been well established that Li⁺ ion is solvated by two TFSA[−] anions acting as a bidentate ligand below the Li⁺ ion mole fraction of $x_{\text{Li}^+} < 0.2$, although the Li⁺ ion tends to form an oligomer species at higher concentration.

In this paper, we have studied the solvation structure of Li⁺ ion in 1-ethyl-3-methylimidazolium bis(fluorosulfonyl)amide [C₂mIm⁺][FSA[−]] ionic liquid solutions dissolving LiFSA salt. Interestingly, we found that the Li⁺ ion solvation in the FSA[−] based ionic liquid was rather different from that in the corresponding TFSA[−] based ones, although both anions have similar molecular structures. To yield further insight into this unusual solvation number, the temperature dependence of the Li⁺ ion solvation number was evaluated. Though the TFSA[−] solvation number is independent of temperature, the Li⁺ ion solvation number in the FSA[−] based ionic liquid evidently decreases with increasing temperature, which suggests that the Li⁺ ion in the FSA[−] based ionic liquid is accompanied by two of more weakly coordinating monodentate FSA[−] together with one of ordinary bidentate FSA[−]. In addition, to yield more direct structural evidence, high-energy X-ray diffraction experiments were performed accompanied by molecular dynamics (MD) simulations.

■ EXPERIMENTAL SECTION

Materials. [C₂mIm⁺][FSA[−]] of electrochemical grade (Dai-ichi Kogyo Seiyaku Co. Ltd.) was dried in a dried vacuum chamber at 323 K for more than 48 h. Water content was checked by a Karl Fischer method to be below 50 ppm. Sample solutions of dissolving LiFSA salt (Dai-ichi Kogyo Seiyaku Co. Ltd.) were prepared in a glovebox (Miwa Mfg. Co., Ltd.) filled with argon gas, in which the water content was kept below 1 ppm.

Raman Spectroscopy. Details of Raman spectra measurements were similar to those for our previous Raman studies.^{56,60,61,63,70} Temperature varying Raman measurements were carried out at 298, 313, 329, 347, and 364 K with a hermetically sealed quartz cell whose temperature fluctuation was kept within ± 0.3 K at a given temperature during the

measurement. The Raman spectra correction employed in this study is as follows. Raw Raman spectra were corrected in terms of solution densities, which had been measured in advance at the respective lithium salt mole fraction and temperature corresponding to those in Raman measurements, and were normalized based on the adequate Raman band arising from the [C₂mIm⁺] cation.⁷⁰ By using the thus corrected Raman intensity $I(\nu)$, Raman spectra are represented as a form of the reduced Raman intensity $R(\nu)$ given as follows:

$$R(\nu) \equiv I(\nu)(\nu_0 - \nu)^{-4} \nu [1 - \exp(-h\nu/kT)] \quad (1)$$

where ν_0 and ν cm^{−1} represent the frequencies of the irradiated laser light and Raman shift, respectively, and the others have their usual meanings.⁴¹ The obtained $R(\nu)$ spectra were deconvoluted into single components of *pseudo*-Voigt functions by nonlinear least squares analyses.

With Raman spectra of ionic liquids dissolving a lithium salt, one may frequently divide solvent molecules (ions) into two approximate kinds: Those bound and those not bound to the metal ion. (Hereafter, we call the former and the latter ions *bound* and *free*, respectively.) Within such an approximation, the solvation number of a metal ion can be evaluated. To yield the solvation number of the lithium ion in the ionic liquid, the following procedure in our previous study^{60,61,63,70} was adopted: the integral intensity I_f of the deconvoluted Raman band of the *free* FSA[−] in the bulk ionic liquid is given as $I_f = J_f c_f$ where J_f and c_f stand for the molar Raman scattering coefficient and the concentration of the *free* FSA[−] in the bulk, respectively. The c_f is given as $c_f = c_t - c_b = c_t - n c_{\text{Li}}$, where c_t and c_b denote the concentrations of total and *bound* FSA[−] (solvated to the lithium ion), respectively, and c_{Li} and n denote the concentration and the solvation number of the lithium ion, respectively. By inserting the equation into $I_f = J_f c_f$ the following relationship was obtained as $I_f/c_{\text{Li}} = J_f(c_t/c_{\text{Li}} - n)$. If the solvation number of the lithium ion in the ionic liquid is kept unchanged under the examined experimental conditions, the plots of I_f/c_{Li} against c_t/c_{Li} should give a straight line; thus the value of n is obtained as $n = -\beta/\alpha$ from a slope $\alpha = J_f$ and an intercept $\beta = -J_f n$.

Ab Initio Calculations. The Gaussian 03 program⁷¹ was used for the ab initio molecular orbital calculations. The basis sets implemented in the Gaussian program were used. The geometries of complexes were fully optimized at the HF/6-311G** level. The intermolecular interaction energies (E_{int}) were calculated at the MP2/6-311G**//HF/6-311G** level by the supermolecule method.^{72,73} Our previous calculations of the [C₂mIm⁺][BF₄[−]] and LiTFSA complexes^{69,74} show that the basis set effects on the calculated interaction energies of the complexes are very small, if basis sets including polarization functions are used, and that the effects of electron correlation beyond MP2 are negligible. Therefore we calculated the interaction energies of the complexes at the MP2/6-311G** level in this work. The basis set superposition error (BSSE)⁷⁵ was corrected for all the interaction energy calculations using the counterpoise method.⁷⁶ The stabilization energy by the formation of complex from isolated molecules (E_{form}) was calculated as the sum of the E_{int} and the deformation energy (E_{def}), which is the sum of the increase of energies of molecules by deformation of the geometries associated with the complex formation.⁶⁹ The E_{def} was calculated at the MP2/6-311G** level.

High-Energy X-ray Diffraction (HEXRD) Experiments. HEXRD measurements were carried out at 298 K using the

BL04B2 beamline of SPring-8 at the Japan Synchrotron Radiation Research Institute (JASRI).^{77,78} Sample ionic liquid was set in a cell consisting of a 2 mm thick polyether ether ketone plate as a body with Kapton films as an X-ray window hermetically sealed with Kalrez O rings, and stainless steel cover plates. Monochrome 61.6 keV X-rays were obtained using a Si(220) monochromator. The observed X-ray intensity was corrected for absorption⁷⁹ and polarization. Incoherent scatterings⁸⁰ were subtracted to obtain coherent scatterings, $I_{\text{coh}}(Q)$. The X-ray structure factor $S^{\text{HEXRD}}(Q)$ and X-ray radial distribution function $G^{\text{HEXRD}}(r)$ per stoichiometric volume were respectively obtained according to

$$S^{\text{HEXRD}}(Q) = \frac{I_{\text{coh}}(Q) - \sum n_i f_i(Q)^2}{(\sum n_i f_i(Q))^2} + 1 \quad (2)$$

$$G^{\text{HEXRD}}(r) - 1 = \frac{1}{2\pi^2 r \rho_0} \int_0^{Q_{\text{max}}} Q \{S(Q) - 1\} \sin(Qr) \exp(-BQ^2) dQ \quad (3)$$

where n_i and $f_i(Q)$ denote the number and the atomic scattering factor of atom i ,⁸¹ respectively, ρ_0 is the number density, and B is the damping factor. All data treatment was carried out using the program KURVLR.⁸²

MD Simulations. All MD simulations in this study are based on the OPLS-AA manner^{83,84} except for combination rules: Lorentz–Berthelot rules (arithmetic and geometric means for σ and ϵ , respectively). CLaP force fields^{85–89} were used for the neat ionic liquids in this study to maintain consistency with our previous work.⁹⁰ Lennard–Jones parameters proposed by Soetens et al. for the lithium ion in nonaqueous cyclic/acyclic carbonate solutions were employed.⁹¹

In our simulations, Gear’s predictor–corrector algorithm^{92,93} was employed for integration of the equations of motion with 0.2 fs time steps. The systems contained 256 ion pairs under NTP ensemble conditions controlled by Nose’s thermostat^{94,95} and the Parrinello–Rahman barostat.^{96,97} The latter was always set to atmospheric pressure. Long-range interactions were estimated using the Ewald summation method with real-space cutoff distances of 11 Å. The simulation runs typically consisted of 2.5–3.5 ns equilibration periods followed by 0.5 ns production runs, whose trajectories were then analyzed. All simulations were carried out using a Fujitsu Materials Explorer 5.0 program suite on the Fujitsu PRIMEQUEST 580 at the Computing and Communications Center, Kyushu University.

RESULTS AND DISCUSSION

Raman Spectroscopy. Typical Raman spectra in the range 200–1600 cm^{-1} at 298 K are shown for $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ ionic liquid dissolved LiFSA salt (0–0.772 mol kg^{-1}) in Figure S1 of the Supporting Information.

Figure 1 shows the Li^+ ion concentration dependence of Raman spectra for $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ containing LiFSA salt in the frequency range 260–400 cm^{-1} , where conformational isomerism of the FSA^- anion is clearly observed.^{54,56} The observed bands at 293, 328, and 360 cm^{-1} can be assigned to the bands at 275, 316, and 350 cm^{-1} predicted for the trans conformer with density functional theory (DFT) calculations at the B3LYP/6-311+G(d) level, respectively, while the bands at 305, 320, and 353 cm^{-1} can be ascribed to the predicted bands at 278, 316, and 344 cm^{-1} , respectively, for the cis conformer.

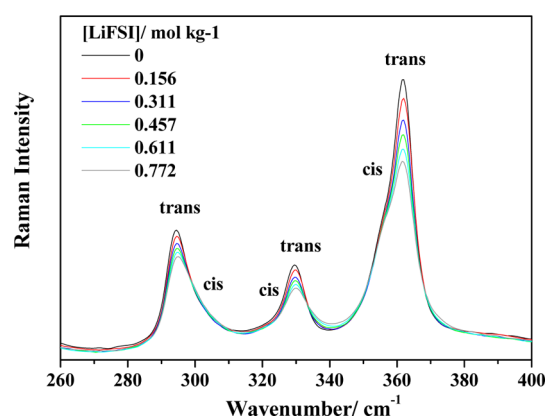


Figure 1. Raman spectra of $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ ionic liquid dissolved LiFSA salt (0–0.772 mol kg^{-1}) in the frequency range 260–400 cm^{-1} at 298 K.

When the Li^+ concentration was increased, the former Raman bands arising from the trans isomer evidently decreased, though no significant peak position shift and no newly appeared peak were found, as shown in Figure 1, indicating that the cis isomer of FSA^- is preferred when FSA^- binds to the Li^+ ion in the FSA^- based ionic liquid. Similar behavior can be found in the TFSA^- solvation to the Li^+ ion^{57–60} and other metal ions.^{80,81} The preferred cis conformation of the solvated TFSA^- can be ascribed to the interaction between the Li^+ ion solvated cluster and ionic liquid composing a cation such as C_2mIm^+ as the second solvation sphere of the Li^+ ion.⁶³ More stabilized FSA^- cis isomer in the first solvation sphere of the Li^+ ion may originate from the same mechanism.

Upon the addition of LiFSA salt, remarkable spectral variations were found in the frequency ranges 680–800 and 1160–1270 cm^{-1} as seen in Figure 2; i.e., new Raman bands of

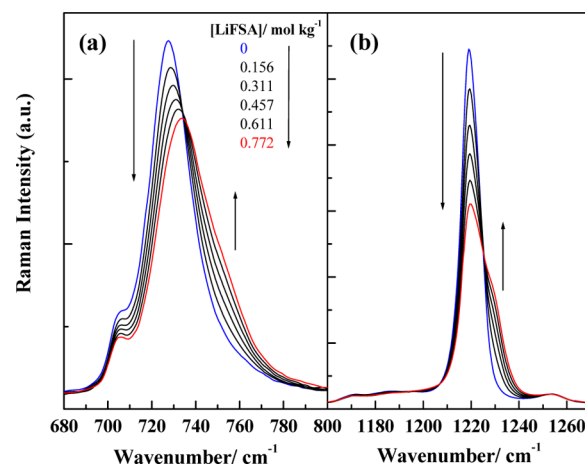


Figure 2. Raman spectra of $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ ionic liquid dissolved LiFSA salt (0–0.772 mol kg^{-1}) in the frequency ranges (a) 680–800 and (b) 1160–1270 cm^{-1} at 298 K.

744 and 1230 cm^{-1} appeared. In these regions, the Raman bands of 731 and 1220 cm^{-1} can be ascribable to the $\nu_s(\text{N-S})$ and the $\nu_s(\text{S-O})$ vibrations of *free* FSA^- anion, respectively.^{54,56} With the increase of the Li^+ ion concentration, these *free* Raman bands were weakened, while the newly appeared bands were intensified. Similar variations were observed in the TFSA^- based ionic liquids; i.e., the Raman band of the $\nu_s(\text{N-})$

S) vibration of TFSA^- around 742 cm^{-1} decreased, while a new band shifted to the higher frequency side appeared with the addition of LiTFSA salt, and increased with increasing lithium salt concentration. It has been well established that the former and latter bands are assigned to the bulk TFSA^- (*free*) and that bound to the Li^+ ion (*bound*), respectively. A similar assignment can be applied to the FSA^- based ionic liquid. It is thus deduced that the newly appeared bands of 744 and 1230 cm^{-1} arise from the FSA^- bound to the Li^+ ion. In the crystal, one of the Li^+ ions is coordinated by three of bidentate FSA^- in regular trigonal prismatic arrangement, while two of the Li^+ ions are coordinated identically by six FSA^- forming distorted octahedrons with a monodentate manner. On the other hand, each FSA^- anion binds to one Li^+ ion with a bidentate manner, at the same time bridging two Li^+ ions acting as a monodentate ligand.^{60,63} Such an FSA^- anion in the crystal yields strong Raman bands of 774 and 1246 cm^{-1} , and the former is more intense. The Raman spectrum of FSA^- anion in LiFSA crystal is thus entirely different from those of the anions solvated to the Li^+ ion in ionic liquid; i.e., peak positions of the bands ascribable to the FSA^- solvated to the Li^+ ion in ionic liquid are 744 and 1230 cm^{-1} , whose intensity is comparable. This clearly indicates that the solvation structure of the Li^+ ion in the FSA^- based ionic liquid is rather different from those found in the crystal. A similar difference can be found in the TFSA^- solvate crystals.^{98–102}

To clarify the Li^+ ion solvation structure in the FSA^- based ionic liquid, the Li^+ ion solvation numbers were evaluated from the observed Raman spectra variations in both ranges $680\text{--}800$ and $1160\text{--}1270\text{ cm}^{-1}$ according to procedures described in the Experimental Section. Raman band deconvolution analyses were satisfactorily performed as shown in Figure S2 in the Supporting Information. As can be seen in Figure S3 in the Supporting Information, the $I_{\text{f}}/m_{\text{M}}$ vs $m_{\text{T}}/m_{\text{M}}$ plots gave respective straight lines, and values of the Li^+ solvation number n were successfully obtained to be $3.0(1)$ and $2.9(1)$ from the 731 and 1220 cm^{-1} bands, respectively. Taking into account the experimental errors, it is suggested that the Li^+ ion is solvated with three FSA^- anions in the $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ ionic liquid at the examined temperature. Interestingly, though the FSA^- anion is analogue on the TFSA^- anion in terms of F atom substitution instead of CF_3 groups, the Li^+ ion solvation number of 3 differs significantly from those values previously reported for the TFSA^- based ionic liquids; i.e., two TFSA^- bind to the Li^+ ion in the TFSA^- ionic liquids in the dilute Li^+ ion concentration range. In addition, according to a recent PGSE NMR study, the FSA^- solvation number of 3 to the Li^+ in the FSA^- based ionic liquids can be adequately explained by its transport properties.⁴⁶ (More recently, analysis of the Stokes–Einstein relation was revisited to obtain a more reasonable value of the friction parameter c in the Stokes–Einstein relationship of $D = kT/c\pi\eta a$ to be 4.4 for FSA^- , while $c = 4.5$ for TFSA^- assumed as the $[\text{Li}(\text{TFSA})_2]^-$ species in the common cation C_2mIm^+ ionic liquids.)

To yield further insight into larger solvation numbers of the Li^+ ion in the FSA^- based ionic liquids compared with the TFSA^- , the temperature dependence of Raman spectra was explored as shown in Figure 3. In Figure 3, the Raman intensity was normalized based on the most intense peak at 1220 cm^{-1} for the neat ionic liquid. As one immediately notices from Figure 3, the respective *free* Raman band intensity (height) for the given Li^+ ion concentration at 364 K is larger than the corresponding ones at 298 K , suggesting that the Li^+ ion

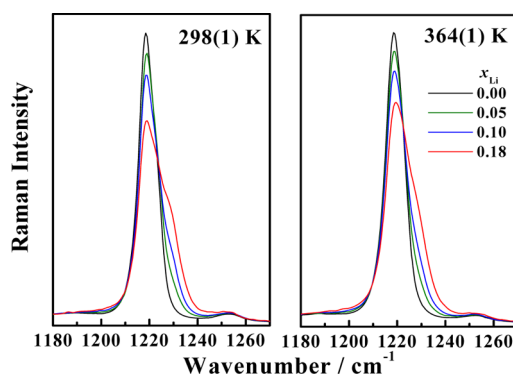


Figure 3. Raman spectra of $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ ionic liquid dissolved LiFSA salt ($0\text{--}0.772\text{ mol kg}^{-1}$) in the frequency range $1160\text{--}1270\text{ cm}^{-1}$ at (left) 298 and (right) 364 K .

solvation number decreases with elevating temperature. In addition, $\Delta\nu$, the value of the higher frequency shift upon coordination to the Li^+ ion, can be a useful sign. As shown in Figure S4 in the Supporting Information, the $\Delta\nu$ for FSA^- solvation became smaller with elevating temperature, while it was practically kept constant for TFSA^- solvation. According to a general rule between a bond length (bond strength) and a coordination number, one may consider that $\Delta\nu$ increased (higher frequency shift) with the decrease of solvation number at higher temperature. Nevertheless, it is the reverse of the case that the temperature dependence of $\Delta\nu$ for FSA^- solvation is an experimental fact. Therefore, a lower frequency shift of the $\Delta\nu$ for FSA^- solvation indicates that Li^+ ions of different coordination numbers and/or manners such as monodentate and bidentate may coexist in the FSA^- based ionic liquid.

To clarify more quantitatively, the Li^+ ion solvation numbers at various temperatures were estimated, and $I_{\text{f}}/m_{\text{M}}$ vs $m_{\text{T}}/m_{\text{M}}$ plots at 364 K and the temperature dependence of the Li^+ solvation numbers are shown in Figure 4 and in Figure S5 in

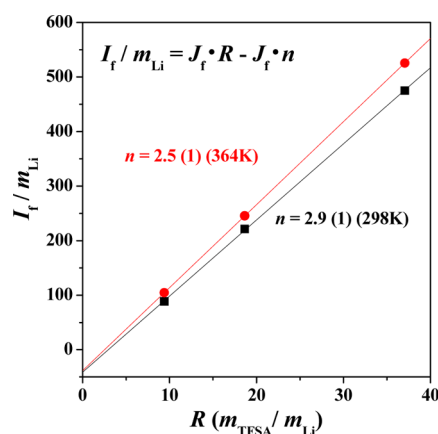
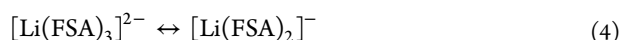


Figure 4. Plots of $I_{\text{f}}/m_{\text{Li}}$ against $m_{\text{TFSA}}/m_{\text{Li}}$ for the 1220 cm^{-1} band at 298 (black) and 364 K (red).

the Supporting Information accompanied by those in the TFSA^- based ionic liquid for comparison. As clearly shown in Figure S5 (Supporting Information), the Li^+ ion solvation numbers at higher temperatures were also successfully determined to be $2.93(6)$, $2.84(9)$, $2.74(6)$, $2.69(8)$, and $2.61(8)$ based on the Raman band at 731 cm^{-1} at 298 , 313 , 329 , 347 , and 364 K , respectively. Similarly, $2.94(6)$, $2.86(9)$, $2.76(5)$, $2.67(5)$, and $2.52(2)$ were obtained as the Li^+ ion

solvation numbers from the Raman band at 1230 cm^{-1} at the respective same temperatures. Clearly, the Li^+ ion solvation number decreases at higher temperature. It should be noted that, in the TFSA[−] based ionic liquids, it has been established that two TFSA[−] anions coordinate to the Li^+ ion with a bidentate manner and that, in fact, the solvation number is practically independent of temperature.⁶³ It is supposed that liberation of a bidentate ligand bound to a metal ion is rather difficult relative to monodentate ones, because, in general, it strongly binds to the metal ion, if no static and/or electrostatic repulsion operates among the solvated ligands. Therefore, the desolvation of FSA[−] anion at higher temperature indicates that at least one FSA[−] anion may coordinate to the Li^+ ion as a monodentate ligand.

For further insight, the temperature dependence of the Li^+ ion solvation number in the FSA[−] based ionic liquid was analyzed. Clearly, the Li^+ ion solvation number of 2.5 at 348 K suggests that Li^+ ions of different solvation numbers are in equilibrium in this temperature range: $[\text{Li}(\text{FSA})_2]^-$ and $[\text{Li}(\text{FSA})_3]^{2-}$. Though rough, we assumed the equilibrium and defined the equilibrium constant as follows:



$$K = \frac{[\text{Li}(\text{FSA})_2]^-}{[\text{Li}(\text{FSA})_3]^{2-}} \quad (5)$$

because the concentration (actually, activity) of the desolvated FSA[−] anion is supposed to be kept constant due to the excess amount of the bulk free FSA[−]. Hence, K can be written as

$$K = \frac{3 - n}{n - 2} \quad (6)$$

where n stands for the average solvation number. The Gibbs free energy change for the equilibrium ΔG° can be written as $\Delta G^\circ = -RT \ln K$, where R and T have the ordinary meanings, and also as $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ by using the enthalpy and entropy changes ΔH° and ΔS° , respectively. Hence, ΔH° and ΔS° can be evaluated from the relationship $-R \ln K = \Delta H^\circ/(1/T) - \Delta S^\circ$.

Figure 5 shows the $-R \ln K$ vs $1/T$ plots with the Raman bands at 731 and 1220 cm^{-1} . Both plots fall on the respective

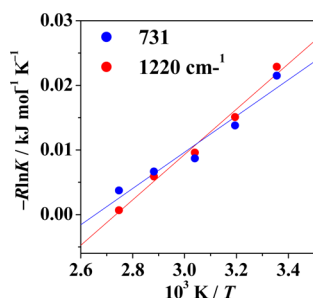


Figure 5. The van't Hoff plots for the equilibrium $[\text{Li}(\text{FSA})_3]^{2-} \leftrightarrow [\text{Li}(\text{FSA})_2]^-$.

straight lines to yield ΔH° and ΔS° adequately. The values of ΔH° and $T\Delta S^\circ$ at 298 K are 28(4) kJ mol^{-1} and 21(3) kJ mol^{-1} from the former and 35(2) kJ mol^{-1} and 29(2) kJ mol^{-1} from the latter Raman bands, respectively. ΔH° and $T\Delta S^\circ$ suggest that the FSA[−] anion loosely bound to the Li^+ ion in the $[\text{Li}(\text{FSA})_3]^{2-}$ desolvates toward the bulk ionic liquid to acquire entropy. Thus, thermodynamic quantities for the FSA[−]

desolvation from the Li^+ ion agree well with the Raman spectroscopic conclusion: one or more monodentate FSA[−] exist in the first solvation shell of the Li^+ ion of $[\text{Li}(\text{FSA})_3]^{2-}$ in the FSA[−] based ionic liquid around room temperature.

Ab Initio Calculations. The optimized geometries for the $[\text{Li}(\text{FSA})]$, $[\text{Li}(\text{FSA})_2]^-$, and $[\text{Li}(\text{FSA})_3]^{2-}$ complexes are shown in Figure 6. The stabilization energies (E_{form}) by the formation of the complexes from isolated ions are also shown in Figure 6.

The $[\text{Li}(\text{FSA})]$ complex has two stable geometries (**1a** and **1b**). The FSA[−] anion is cis form in **1a** and trans form in **1b**. **1a** is 9.2 kJ mol^{-1} more stable than **1b**. The three stable geometries for the $[\text{Li}(\text{FSA})_2]^-$ complex (**2a–2c**) are shown in Figure 6. The conformations of the two FSA[−] anions in **2a**, **2b**, and **2c** are cis–cis, trans–cis, and trans–trans, respectively. The order of the stability is **2a** > **2b** > **2c**. The calculated energies of **2b** and **2c** relative to **2a** are 6.5 and 12.6 kJ mol^{-1} , respectively. The cis form bidentate FSA[−] anions coordinate with Li^+ ion in the most stable geometries of the $[\text{Li}(\text{FSA})]$ and $[\text{Li}(\text{FSA})_2]^-$ complexes (**1a** and **2a**), which shows that the FSA[−] anion prefers to have the cis form, when bidentate FSA[−] anion coordinates with Li^+ ion.

Thirty-eight stable geometries were obtained for the $[\text{Li}(\text{FSA})_3]^{2-}$ complex by geometry optimizations. They can be classified into four groups. (1) One bidentate and two monodentate FSA[−] anions coordinate with Li^+ ion. (2) Three monodentate FSA[−] anions coordinate with Li^+ ion. (3) Two bidentate and one monodentate FSA[−] anions coordinate with Li^+ ion. (4) Three bidentate FSA[−] anions coordinate with Li^+ ion. The most stable geometries in groups 1–4 (**3a–3d**) are shown in Figure 6. The order of the stability is **3a** > **3b** > **3c** > **3d**. The calculated energies of **3b**, **3c**, and **3d** relative to **3a** are 11.2, 12.3, and 23.2 kJ mol^{-1} , respectively. The numbers of oxygen atoms of FSA[−] anions which have contact with the Li^+ ion (coordination number) are 4, 3, 5, and 6 in groups 1, 2, 3, and 4, respectively. The coordination number of the most stable geometry **3a** is 4, which is the same as the coordination number in the stable geometry of the Li^+ complex with TFSA[−] anions ($[\text{Li}(\text{TFSA})_2]^-$). The bidentate FSA[−] anions in the most stable geometries for the four groups (**3a–3d**) have the cis form as in the cases of the most stable geometries of the $[\text{Li}(\text{FSA})]$ and $[\text{Li}(\text{FSA})_2]^-$ complexes (**1a** and **2a**), while the monodentate FSA[−] anions have the trans form in **3a–3c**.

The average Li^+ –O distance in the $[\text{Li}(\text{FSA})_3]^{2-}$ complex is longer than that in the $[\text{Li}(\text{TFSA})_2]^-$ complex. The Li^+ –O distances for bidentate FSA[−] anion in **3a** are 2.02 Å, while the distances for monodentate FSA[−] anions are 1.92 Å. The average Li^+ –O distance is 1.97 Å. The average Li^+ –O distance calculated for the $[\text{Li}(\text{TFSA})_2]^-$ complex at the same level is 1.94 Å.

The interaction of the Li^+ with bidentate FSA[−] anion is stronger than that with monodentate FSA[−] anion. The interaction energy of Li^+ with each FSA[−] anion was calculated using the geometry of **3a**. The calculated interaction energy (E_{int}) with bidentate FSA[−] anion is $-560.6 \text{ kJ mol}^{-1}$, while that with monodentate anion is $-444.3 \text{ kJ mol}^{-1}$.

The monodentate FSA[−] anions in **3a** have large conformational flexibility due to the internal rotation of the Li^+ –O, while the conformational flexibility of the bidentate FSA[−] anions in **2a** is small. The release of a FSA[−] anion from the $[\text{Li}(\text{FSA})_3]^{2-}$ complex will decrease the entropy of the internal rotation of the complex, which is one of the causes of the small experimental

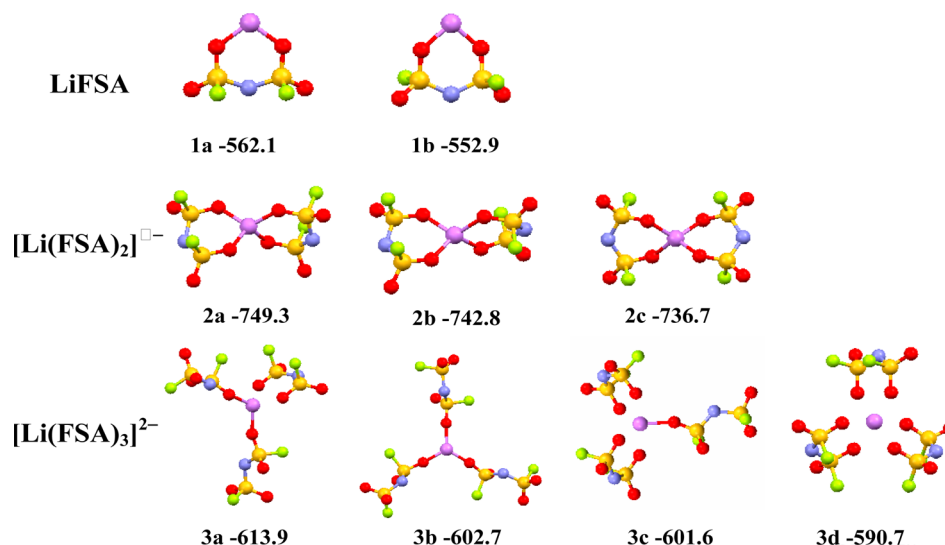


Figure 6. Optimized geometries and interaction energies E_{int} (kJ mol⁻¹) for $[\text{Li}(\text{FSA})_n]^{(1-n)+}$ ($n = 1, 2, \text{ and } 3$).

ΔS° associated with the liberation of a FSA^- anion from the $[\text{Li}(\text{FSA})_3]^{2-}$ complex.

HEXRD Experiments and MD Simulations. In general, it is quite difficult to elucidate the Li^+ ion solvation structure in solution by means of an X-ray scattering technique because of rather small X-ray scattering ability. However, we recently found that the peak ascribable to the $\text{Li}^+\text{--O}$ (TFSA⁻) of $[\text{Li}(\text{TFSA})_2]^-$ in $[\text{C}_2\text{mIm}^+][\text{TFSA}^-]$ appeared at about 1.9 Å with HEXRD experiments of high real-space resolution. We thus attempted to perform HEXRD experiments to find the corresponding peak arising from the $[\text{Li}(\text{FSA})_3]^{2-}$. X-ray structure factors $S(Q)$ in the Q range up to 25 Å⁻¹ are shown in Figure S6 in the Supporting Information. $S(Q)$ for neat $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ is in good agreement with that previously reported;⁵⁵ a peak at 1.48 Å⁻¹ with a shoulder of 0.98 Å⁻¹ appeared in the Q range below 2.0 Å⁻¹. These are characteristic for the intermolecular liquid structure of the ionic liquid, so that details of neat $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ will be discussed in our future work. However, it is worth pointing out that the intensities at 1.48 and 0.98 Å⁻¹ increased and decreased, respectively, with increasing Li^+ ion concentration, suggesting that the density fluctuation should change in the long range of ~10 Å beyond the first solvation shell of the Li^+ ion.

Radial distribution functions $G(r)$ for $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ dissolved LiFSA are shown in Figure 7. Sharp peaks below 4.0 Å—1.5, 2.2, 2.4, and 2.7 Å—can be assigned to the intramolecular atom–atom correlations. As can be seen in the inset of Figure 7(top), a small but significant peak at about 1.9 Å⁻¹ was intensified with increasing Li^+ concentration. Similarly, with the $[\text{C}_2\text{mIm}^+][\text{TFSA}^-]$ case, the peak can be ascribable to the $\text{Li}^+\text{--O}(\text{FSA}^-)$ atom–atom correlation in the first solvation shell of the Li^+ ion in $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$. In addition, a rather small, but noticeable difference was found in the distance of the $\text{Li}^+\text{--O}$ correlations: the $\text{Li}^+\text{--O}(\text{FSA}^-)$ of 1.94 Å is slightly longer than the $\text{Li}^+\text{--O}(\text{TFSA}^-)$ of 1.86 Å.¹⁰² The difference of 0.08 Å in distance can be significant for the Li^+ ion solvation structure in the respective ionic liquid. The longer $\text{Li}^+\text{--O}(\text{FSA}^-)$ correlation of 1.94 Å is consistent with the larger solvation number of the Li^+ ion in $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ relative to that in $[\text{C}_2\text{mIm}^+][\text{TFSA}^-]$, which agrees with the Raman spectroscopic evidence.

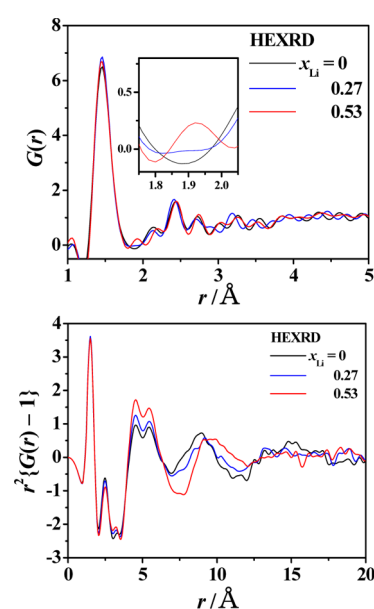


Figure 7. (top) X-ray radial distribution functions $G(r)$ and (bottom) difference radial distribution functions in the form $r^2\{G(r) - 1\}$ for $[\text{Li}^+]_{x_{\text{Li}}}[\text{C}_2\text{mIm}^+]_{(1-x_{\text{Li}})}[\text{FSA}^-]$ ($x_{\text{Li}} = 0.0, 0.27, \text{ and } 0.53$).

The difference radial distribution functions in the form $r^2\{G(r) - 1\}$ for $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$ dissolved LiFSA are also shown in Figure 7. Broad peaks of about 5 Å, which splits into two sharp peaks of 4.5 and 5.5 Å, 9 Å, and 15 Å can be attributable to the atom–atom correlations in the intermolecular interaction. Besides the first neighboring cation–anion, the ions of the same sign and the second layer of the cation–anion can be predominantly ascribable to the respective broad peak. As mentioned above, the long-range density fluctuation clearly changes up to 13 Å by the addition of LiFSA salt; the peak intensity at 5 Å increased without the peak position shift, and the intensities around 8 and 11 Å decreased and increased, respectively, with increasing Li^+ concentration. Similar variations of the long-range density fluctuation have been observed for the $[\text{C}_2\text{mIm}^+][\text{TFSA}^-]$ system, and were assigned to the long-range anion–anion correlations by producing the large Li^+ ion solvated cluster of $[\text{Li}(\text{TFSA})_2]^-$. Thus, the

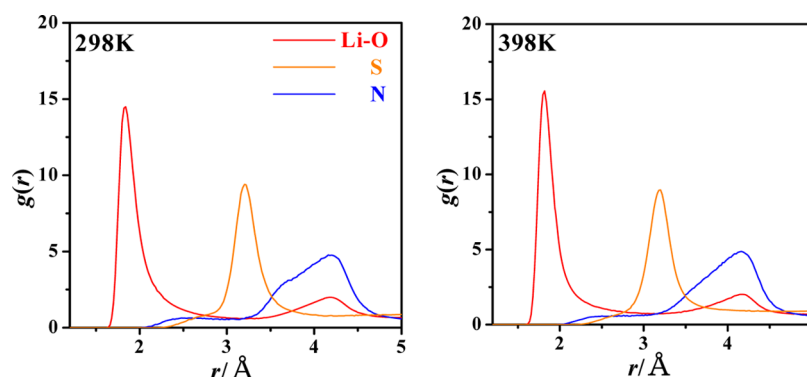


Figure 8. Pair distribution functions $g(r)$ for $[\text{Li}^+]_{x_{\text{Li}}}[\text{C}_2\text{mIm}^+]_{(1-x_{\text{Li}})}[\text{FSA}^-]$ ($x_{\text{Li}} = 0.27$).

variations in the long-range density fluctuation found in this study can be ascribed to the changes in the long-range $\text{FSA}^- - \text{FSA}^-$ correlations caused by the formation of the Li^+ ion solvate cluster of $[\text{Li}(\text{FSA})_3]^{2-}$.

MD simulations were carried out at the same conditions as the HEXRD experiments.

Figure 8 displays pair correlation functions of the Li^+ and O, N, and S atoms of FSA^- ion at 298 K accompanied by those at the higher temperature of 398 K. As can be seen in Figure 8, the most closed $\text{Li}^+ - \text{O}$ peak appeared at 1.84 Å, with the integral value of 3.72 up to 3.0 Å, indicating that the Li^+ ion is mainly tetrahedrally four-coordinated structure. In addition, 3.98 sulfur and 3.09 nitrogen atoms locate at 3.20 Å and about 4 Å, respectively. It should be noted that two kinds of nitrogen exist at 3.72 and 4.19 Å, suggesting that the former and the latter can be attributable to FSA^- anions bound to the Li^+ ion with the bi- and monodentate manners. Moreover, the coordination number of the monodentate FSA^- is larger than that of the bidentate one; the ratio of the monodentate to the other is 1.97. All of the pair correlation functions results suggests two of the monodentate and one of the bidentate FSA^- anions surround the Li^+ ion in $[\text{C}_2\text{mIm}^+][\text{FSA}^-]$.

A typical Li^+ ion solvated cluster of $[\text{Li}(\text{mono-FSA})_2(\text{bi-FSA})]^{2-}$ found in the MD snapshots is shown in Figure 9.

At higher temperature, all of the coordination numbers of oxygen, sulfur, and nitrogen atoms were slightly reduced to 3.44, 3.71 and 2.91, respectively, without significant peak position shifts. This strongly supports Raman experiments.

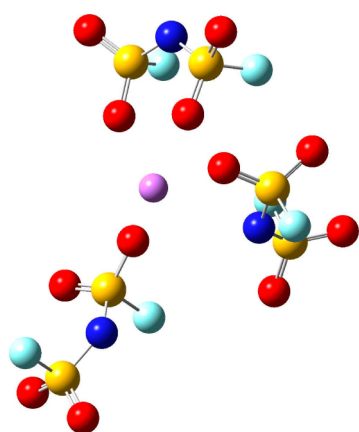


Figure 9. Typical Li^+ ion solvation structure extracted from the MD snapshot.

Interestingly, the first peak of the $\text{Li}^+ - \text{O}(\text{FSA}^-)$ was more sharpened at higher temperature, indicating that fewer Li^+ ion solvated clusters exist at higher temperature relative to an ambient one. Besides, the ratio of the monodentate FSA^- to the bidentate one evaluated from the peak of the $\text{Li}^+ - \text{N}(\text{FSA}^-)$ also slightly decreased to 1.92. MD simulations at higher temperature suggest that the monodentate FSA^- anion preferentially desolvates from the first solvation shell of the Li^+ ion, which also supports Raman spectroscopic results.

CONCLUSION AND THE FUTURE

It was revealed that Li^+ ion in the FSA^- based ionic liquids is solvated with three FSA^- anions by Raman spectra. The fact is rather different from that for the Li^+ ion solvation in the TFSA^- based ionic liquids; i.e., the TFSA^- solvation number is 2 and is independent of temperature. In addition, the solvation number of FSA^- anion decreased to 2 with elevating temperature, which is also different from the solvation in the TFSA^- based ionic liquids. The difference in the Li^+ ion solvation is not only the solvation number but also the coordination manner. The solvated FSA^- anions around the Li^+ ion act as both bidentate and monodentate ligands in the FSA^- system, whereas they act only as a bidentate ligand in the TFSA^- system. Moreover, HEXRD experiments show the peak at about 1.94 Å ascribable to the $\text{Li}^+ - \text{O}(\text{FSA}^-)$ atom–atom correlation, which is evidently longer than that of the $\text{Li}^+ - \text{O}(\text{TFSA}^-)$ correlation of 1.86 Å. The longer binding length of $\text{Li}^+ - \text{O}(\text{FSA}^-)$ suggests the larger solvation number to the Li^+ ion in the FSA^- based ionic liquids. MD simulations at least qualitatively reproduced Raman and HEXRD experimental evidence. However, the detailed structure of the $[\text{Li}(\text{FSA})_3]$ complex, i.e., the population of bi- and monodentate FSA^- anions in the solvation sphere, is not found yet at the present stage. Therefore, neutron scattering experiments with the ^6Li isotopic substitution technique are needed to elucidate the Li^+ ion solvation local structure at an atomistic level.

ASSOCIATED CONTENT

Supporting Information

Additional data of Raman spectra, HEXRD experiments, and MD simulations. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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